

# Md Imran Khan

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imransmailbox@gmail.com | 607-242-5532 | kxanimran.com | linkedin.com/in/imrankahn

## Professional Summary

**Applied Scientist** with a Ph.D. background in optics and computational modeling, specializing in imaging systems, physics-driven algorithms, and data-intensive analysis. Brings 4 years of industry experience in CD-SEM metrology at Intel, combined with deep research experience in experimental optics and numerical simulation. Proven ability to translate physical models into scalable Python workflows for image quality assessment, automation, and tool health monitoring. Actively transitioning toward AI and computer vision-focused roles at the intersection of physics-based modeling, metrology, and applied machine learning.

## Technical Skills

- **Programming & AI:** Python (PyTorch, OpenCV, scikit-image, Pillow, NumPy, pandas, Matplotlib), SQL, Git, Jupyter
- **Data Analysis & Modeling:** Statistical Process Control (SPC), JMP, Statistical Modeling, Dashboarding, Automation, Origin, MATLAB, LabView
- **Metrology & Imaging:** Scanning Electron Microscopy (SEM), CD-SEM, Optical Microscopy, Image Quality Analysis (CNR, Sharpness, FFT, GLCM), TRPL (Time Resolved Photoluminescence)
- **Fabrication & Tools:** Clean Room, Sputtering, Photolithography, DLS

## Experience

### **Metrological Process Engineer (CD-SEM) at Intel Corporation**

Intel Corporation, Hillsboro, USA | November 2021 – August 2025

- Optimized fleet-wide measurement stability for 8 high-precision CD-SEM systems in a 24/7 high-volume manufacturing environment. Leveraged Statistical Process Control (SPC) and automated calibration logic to minimize measurement drift and maximize yield.
- Engineered an automated anomaly detection system using Python to monitor high-dimensional sensor data (beam parameters, sub-system trends). Transitioned the team from reactive troubleshooting to proactive system optimization through real-time data visualization.
- Automated multi-variate statistical pipelines to evaluate tool-to-tool matching and measurement precision for DCCD process. Reduced data processing time by 75% (from ~5 hrs to less than 1 hour) while improving reporting accuracy for fleet-wide consistency checks.
- Developed Python workflows for processing FCCD datasets and integrating results into long-term SPC systems.
- Collaborated with cross-functional team for SEM metrology stability, coordinating with hardware and software integration teams to troubleshoot complex electron-optical drift and optimize image acquisition.

## Education

- Ph.D., Physics (Optics & Computational Modeling), UC Merced | August 2016 – October 2021
- M.S., Physics, Binghamton University (SUNY) | May 2016

## Research Experience

### Graduate Research Scholar

University of California, Merced, California. | Aug, 2016 - Oct, 2021

### Computational Modeling & Nano-fabrication of Plasmonic Metastructures:

- Developed a hybrid full-wave modeling framework to study angular scattering, extinction, anisotropy, and broadband cloaking in plasmonic core-shell systems.
- Combining the Method of Fundamental Solutions (MFS) and Foldy-Lax multiple-scattering theory to simulate scattering from dielectric cores coated with plasmonic nanoparticle assemblies.
- Implemented Python and MATLAB solvers for large-scale parameter sweeps (particle size, filling fraction, geometry, wavelengths)
- Nano-fabricated core-shell plasmonic structures, characterized geometry using SEM, and correlated experimental scattering measurements with simulation predictions.
- Demonstrated strong agreement between modeled and measured optical response

### Optical Metrology for Energy Transfer in Nanoclusters

- Designing and aligning a high-stability time-resolved photoluminescence (TRPL) measurement system optimized for dilute samples and weak emitters
- Quantified photoluminescence lifetime changes and intensity quenching to distinguish Förster (FRET) and nanoparticle-mediated (NSET) energy-transfer regimes in gold nanocluster systems (AuNCs).

## AI & Computer Vision Projects(selected)

- **Chest X-Ray Pneumonia Classification:** End-to-end CV pipeline from PCA + SVM to multiclass ResNet models with Grad-CAM interpretability.
- **Automated SEM Image Quality Assessment:** Computer vision toolkit for sharpness, contrast, FFT, and texture-based SEM image metrics.
- **California Housing Price Prediction:** Comparative ML workflow with feature engineering and model benchmarking (XGBoost, Random Forests).

## Publications (Selected)

- Khan, M.I., et al., "Scattering by nanoplasmonic mesoscale assemblies," JOSA A 42, 1244–1253 (2025).
- Khan, M.I., et al., "Modeling broadband cloaking using 3D nano-assembled plasmonic meta-structures", Optics Express, 2020.

## Awards & Affiliations

- NSF-CREST Graduate Scholar, CCBM, UC Merced
- Graduate Scholar Mentor – Mentored undergraduate researchers on experimental methods and graduate school preparation